

# Key Observations Report

E-Commerce Orders Dataset  
*Exploratory Data Analysis (EDA) — Project 2*

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## 1. Problem Statement

The core question being analyzed is: What patterns and trends exist in e-commerce order data? The analysis focuses on understanding pricing behavior, spending distributions, outliers, relationships between key numerical variables, and customer acquisition sources across 1,200 transactions.

## 2. Methodology

The dataset contains 1,200 order records. The following analytical steps were performed:

- Descriptive statistics calculated for all numerical columns: Quantity, UnitPrice, ItemsInCart, and TotalPrice
- Metrics computed: Count, Mean, Median, Minimum, Maximum, Q1, Q3, and IQR
- Outlier detection applied to TotalPrice using the IQR method
- Correlation analysis conducted between three key variable pairs
- Referral source analysis using COUNTIF to count orders by customer acquisition channel

## 3. Descriptive Statistics Summary

| Metric         | Quantity | Unit Price | Items In Cart | Total Price |
|----------------|----------|------------|---------------|-------------|
| Count          | 1200     | 1200       | 1200          | 1200        |
| Mean           | 2.95     | \$356.41   | 5.49          | \$1,053.97  |
| Median         | 3        | \$364.21   | 5             | \$823.62    |
| Minimum        | 1        | \$11.39    | 1             | \$11.39     |
| Quartile 1     | 2        | \$186.06   | 4             | \$410.52    |
| Quartile 3     | 4        | \$521.57   | 7             | \$1,578.48  |
| Maximum        | 5        | \$699.93   | 10            | \$3,456.40  |
| IQR            | 2        | \$335.51   | 3             | \$1,167.95  |
| Upper Bound    | -        | -          | -             | \$3,330.41  |
| Total Outliers | -        | -          | -             | 8           |

## Correlation Analysis

Pearson correlation was calculated between the following variable pairs:

| Variable Pair              | Correlation | Interpretation                |
|----------------------------|-------------|-------------------------------|
| UnitPrice vs. TotalPrice   | 0.717       | Strong positive correlation   |
| ItemsInCart vs. TotalPrice | 0.393       | Moderate positive correlation |
| UnitPrice vs. Quantity     | 0.015       | Almost zero correlation       |

## Referral Source Analysis

The following table shows the number of orders acquired through each customer referral channel:

| Referral Source  | Order Count | Percentage   |
|------------------|-------------|--------------|
| <b>Instagram</b> | <b>259</b>  | <b>21.6%</b> |
| Email            | 250         | 20.8%        |
| Google           | 241         | 20.1%        |
| Facebook         | 228         | 19.0%        |
| Referral         | 222         | 18.5%        |

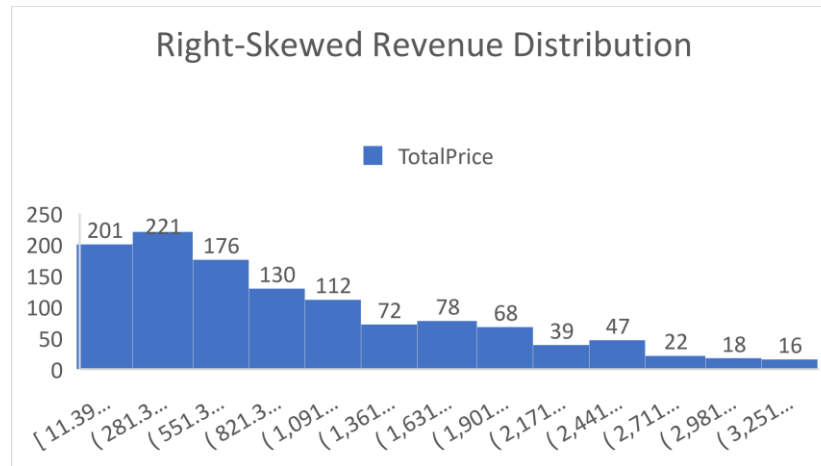
## 4. Key Findings

### Finding 1 — Average order value is higher than the typical order value

The mean TotalPrice is \$1,053.97 while the median is \$823.62. Since the mean is noticeably higher than the median, the data is right-skewed — meaning a few large orders are pulling the average up. The median is the more accurate representation of a typical order.

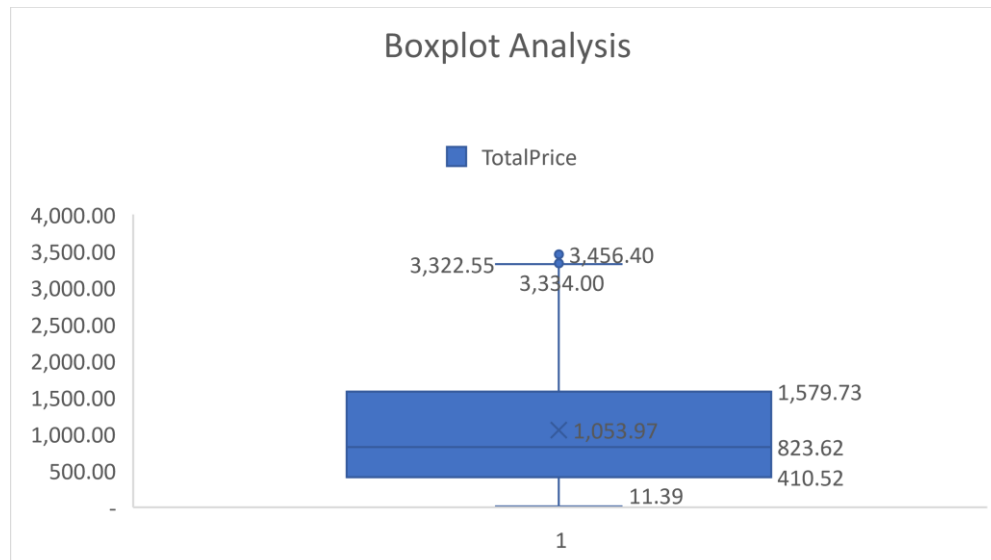
### Finding 2 — TotalPrice has a wide spread

TotalPrice ranges from a minimum of \$11.39 to a maximum of \$3,456.40, with a large IQR of \$1,167.95 (Q1: \$410.52, Q3: \$1,578.48). This shows customers have very different spending levels — some buy very cheaply while others spend significantly more.



### Finding 3 — 8 High-Value Outliers were identified

Using the IQR method, the upper bound was calculated as \$3,330.41. A total of 8 orders exceeded this threshold and were flagged as High Value. These are genuine high-spending customers worth further attention.



Box plot of TotalPrice showing distribution from \$11.39 to \$3,456.40. The dot above the upper whisker represents high-value outlier orders exceeding the IQR upper bound of \$3,330.41, consistent with the 8 outliers identified in the analysis.

#### **Finding 4 — UnitPrice is the strongest driver of TotalPrice**

The correlation between UnitPrice and TotalPrice is 0.717 (strong positive). This means the price of the product is the biggest factor in how much a customer ends up spending. Higher-priced products directly lead to higher total order values.

#### **Finding 5 — Cart size has a moderate influence on spending**

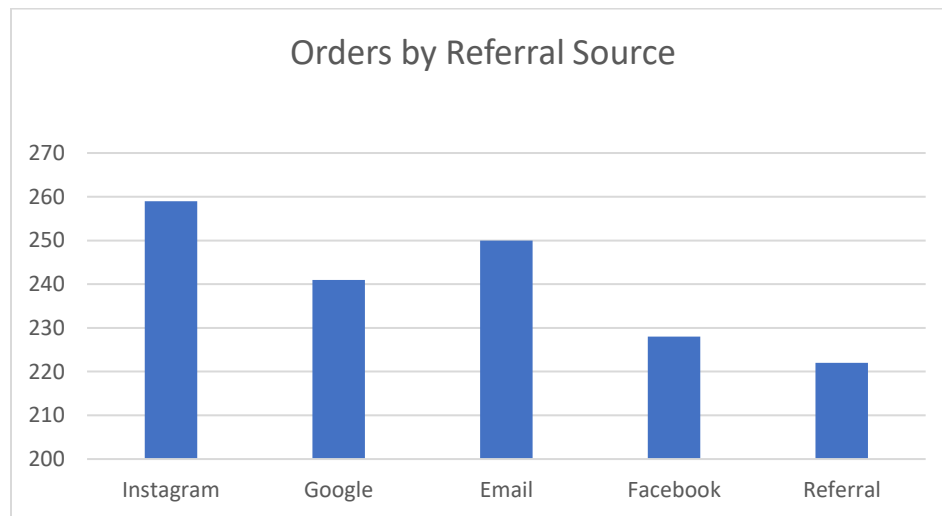
ItemsInCart vs TotalPrice shows a moderate positive correlation of 0.393. Customers with more items in their cart do spend more, but this relationship is not strong enough to be the primary revenue driver.

#### **Finding 6 — Quantity ordered has almost no relationship with UnitPrice**

The correlation between UnitPrice and Quantity is nearly zero (0.015). Customers buy expensive and cheap products in similar quantities — there is no bulk-buying pattern observed for cheaper items.

#### **Finding 7 — Instagram is the top customer acquisition channel**

Instagram drives the highest number of orders at 259, slightly ahead of Email (250) and Google (241). Facebook (228) and Referral (222) trail behind. All five channels are fairly balanced, but Instagram holds a slight edge as the leading source of customer acquisition.



## 5. Recommendations

1. **Use median (\$823.62) rather than mean for reporting typical order value** — the mean is distorted by outliers and gives a misleading picture.
2. **Focus on selling higher-priced products** — since UnitPrice is the strongest revenue driver ( $r = 0.717$ ), promoting premium items will have the most impact on revenue.
3. **Investigate the 8 High-Value outlier orders** — these customers spent well above average and could be VIP buyers worth targeting with loyalty offers.
4. **Do not over-invest in cart-quantity strategies** — since cart size only moderately affects spend ( $r = 0.393$ ), product pricing matters far more than adding more items to the cart.
5. **Prioritize Instagram as the primary marketing channel** — with the highest order count of 259, increasing investment in Instagram content and ads is likely to generate the best customer acquisition return.